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ISPRS International Journal of Geo-Information (2023)

# Can ChatGPT Serve as a "Map Design Brain"?

isprs International Journal of  
Geo-Information



Article

## Mapping with ChatGPT

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CORE ISSUE

LLM-Driven Mapping: Barriers & Limits



## Prompt Input

Natural language  
description  
Define theme & data



## ChatGPT Generation

Understand spatial intent  
Generate Python code

Critical Bottleneck



## IDE Execution

Manual code running  
Google Colab environment



## Map Output

Iterative optimization  
Final visualization



## Thematic Maps

Choropleth Map: US Population Core Comparison

Batch Mapping: County-level maps for southern states

Web Map: Interactive Folium-based visualization



## Mental Maps

Spatial Cognition: Shape, topology, distance

Complex Scenario: Campus layout & functional zones

Fuzzy Reasoning: "5-min walk"  $\approx$  400 meters

NOT End-to-End: Requires Manual Copying & Debugging



# Key Evidence: Thematic Map Comparison (Fig 2 vs 3)



✗ Figure 2: Initial Output

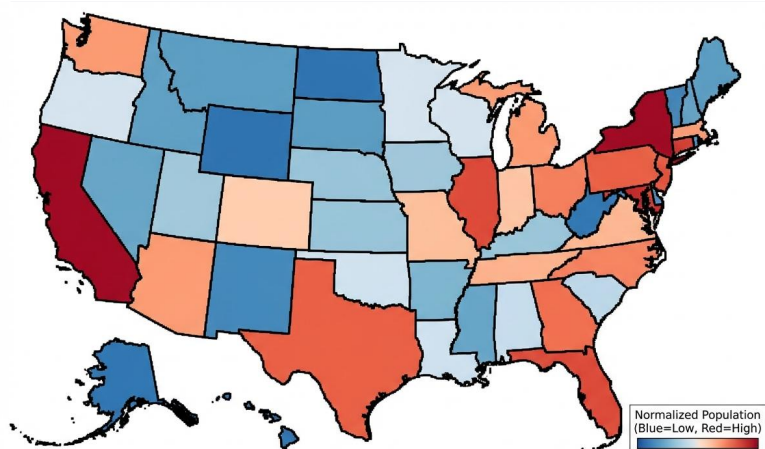


Figure 2: Initial US population choropleth map with problems  
Figure 2 illustrates initial attempts with normalized population data, highlighting projection errors, an unsuitable diverging color scheme, and the incorrect placement of non-contiguous states.

Layout Error: Alaska & Hawaii forcibly attached

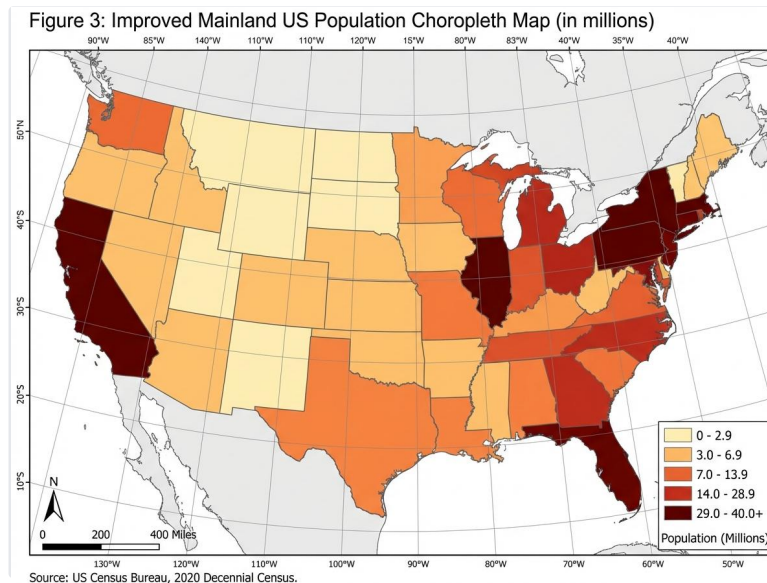
Distortion: Poor projection choice, skewed scale

Misleading Color: Diverging (Blue-Red) inappropriate

Auto-Norm: Data normalized without request

POOR QUALITY

✓ Figure 3: Improved Output



Mainland Focus: Removed incorrect insets

Spatial Ref: Added graticule grid for readability

Logical Color: Sequential scheme (Light-to-Dark)

Standard Class: Natural Jenks, rounded legend

SIGNIFICANTLY IMPROVED



Core Contradiction: Not "One-Click Generation"

ChatGPT cannot independently produce professional maps; the improvement process requires professional cartographic knowledge to identify errors and guide debugging.



# Core Findings: 4 Advantages vs 4 Limitations



## 👍 4 Key Advantages



### Significantly Lowers Barriers

Enables mapping via natural language without coding skills; detailed comments aid learning.



### Boosts Production Efficiency

Supports batch mapping, reusable code frameworks, auto-data sourcing, and context memory.



### Spatial Thinking Capability

Understands shapes, distances, and topology to convert text descriptions into mental maps.



### Inspires Creative Design

Suggests graticules, color schemes, and new tools (e.g., Folium) to enhance map aesthetics.



## 4 Critical Limitations



### Text-Only Input/Output

Cannot process spatial files directly; relies on external tools/code, not end-to-end automation.



### High External Dependency

Constrained by Python library limits, IDE environment, internet access, and data availability.



### Unstable Initial Quality

First outputs often contain errors (projections, colors); requires multiple iterations to fix.



### Unequal User Benefits

Programmers benefit more; beginners struggle with code debugging and error correction.



Core Contradiction: Neither fully automated (requires human intervention) nor quality-guaranteed (initial errors common)



# Critical Conclusion & GIS Significance



## Reproducibility

Random outputs yield inconsistent code, violating scientific reproducibility principles.



## Data Timeliness

Training data ending in 2021 leads to outdated data sources or API recommendations.



## Workflow Value

Is manual debugging faster than regeneration? Does this lower barriers or add new burdens?



## Irreplaceability

Professional skills like projection choice and symbology cannot be fully replaced by AI.



## Ethics & Liability

Who is responsible for AI-generated errors (e.g., wrong borders)? Lack of accountability.



## CORE POSITIONING

ChatGPT should be a **"Co-pilot / Assistive Tool"**, NOT a **"Replacement"**



## Educational Impact

Shift focus from "software operations" to "design principles" and "critical thinking".



## Research Access

Lower visualization barriers, enabling non-GIS scholars to map spatial data easily.



## Future Directions

Integrate with image AI (DALL·E) for "autonomous agents" and build quality control systems.

## SPEAKER'S PROFILE

### Research Interests

Human mobility, Environmental exposure (heatwave, pollution compound exposure), GeoFoundation Model, Geographic representation learning, Geospatial Data Science

**Ad** I am looking for collaborators interested in GeoAI, Spatial Analysis, and their applications in urban science and environmental interactions. My goal is to develop tools and methodologies to empower researchers in exploring geospatial data science. If you are interested in collaborating, please feel free to reach out via email.

**Email:** [jjale.guo@mail.polimi.it](mailto:jjale.guo@mail.polimi.it) **Website:** <https://guojialegeographer.github.io/presentations.html>